

Mathematics for Machine Learning: Homework 4

1. Find the geometric and algebraic multiplicity of the eigenvalue(s) of the matrix $A = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$.

2. Let $A \in \mathbb{R}^{3 \times 3}$ be a matrix such that

$$A\mathbf{v}_1 = 3\mathbf{v}_1, \quad A\mathbf{v}_2 = 5\mathbf{v}_2, \quad A\mathbf{v}_3 = -\mathbf{v}_3,$$

for some non-zero vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^3$. Determine the eigenvalues, characteristic polynomial, determinant and the trace of the matrix A . Find the characteristic polynomial of A^T .

3. Compute the trace of the matrix $ABCB^{-1}A^{-1}$ given that $A, B, C \in \mathbb{R}^{5 \times 5}$ with $C\mathbf{v}_i = 2^i\mathbf{v}_i, i = 1, \dots, 5$ for some non-zero vectors $\mathbf{v}_1, \dots, \mathbf{v}_5 \in \mathbb{R}^5$.
4. Find the Cholesky decomposition of the 3×3 matrix

$$A = \begin{bmatrix} 1 & -2 & -3 \\ -2 & 8 & 12 \\ -3 & 12 & 27 \end{bmatrix}.$$

5. Compute the eigendecomposition of a (symmetric) matrix

$$A = \begin{bmatrix} 0 & 4 \\ 4 & 6 \end{bmatrix}.$$

6. Find the SVD of the matrix

a) $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix},$

b) $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}.$

7. Using the definition of limit, prove that

a) $\lim_{n \rightarrow \infty} \frac{2 + (-1)^n}{n} = 0,$

b) $\lim_{n \rightarrow \infty} \frac{3n \sin n + 1}{2n^2 + 2n - 1} = 0,$

c) $\lim_{n \rightarrow \infty} \frac{2n^2 + 1}{8n^2 - 2n + 10} = \frac{1}{4},$

d) $\lim_{n \rightarrow \infty} \frac{\sqrt{3n^2 + 1}}{\sqrt{n^2 + 2n + 10}} = \sqrt{3}.$

8. Prove that the sequence x_n is divergent

a) $x_n = \frac{n}{n+1} \cos \frac{2\pi n}{3},$

c) $x_n = \frac{n^2 - 2n}{n+1},$

b) $x_n = 2^{(-1)^n n},$

d) $x_n = n^2 \sin \frac{\pi n}{4}.$